

Efficiency Through Politeness: Gamification and Fuzzy Logic in Human–Robot Interaction

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Abstract
This project explores how prioritising politeness over efficiency can influence human-robot interaction while under the pressure of a time limit. A gamified, voice-controlled artefact utilises fuzzy logic to gauge politeness using speech pacing and linguistic cues. Calmer and more polite utterances are rewarded through a proportional scoring system. Observations show users adapting their speech patterns when given real-time feedback, with efficiency emerging through higher scores.

Introduction
Human-robot interaction under time constraints often prioritises efficiency. This aligns with Gricean maxims [1] efficient speech can be perceived as blunt or impolite, which can reduce social cooperation.

This project explores a gamified human-robot interaction that deliberately reinforces politeness over efficiency (under English conversational norms). Users will verbally ask the robot to move a ball around a triangular circuit under a time limit. Polite and calm speech is gauged, and points will be given through a proportional scoring system. This reflects the role of politeness in helping social cooperation [2].

This study examines how politeness reinforcement can help build better cooperation in time-constrained interactions between robots and humans.

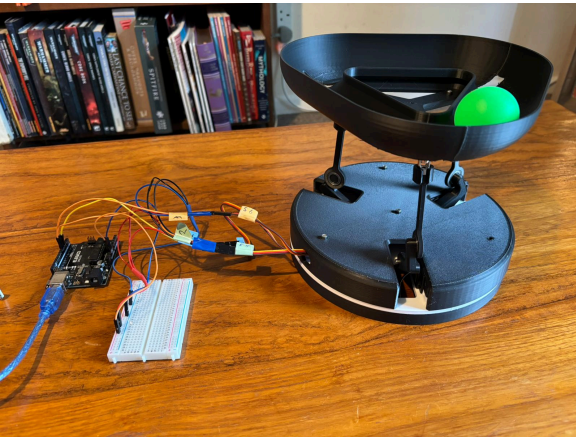


Figure 1: Game interface showing ball position and politeness-based score feedback during interaction.

Interaction
The user interacts with the artefact via a start button and communicates using a microphone. The speech is recognised using VOSK and analysed for perceived calmness and politeness through a fuzzy logic system, using indirect speech act (ISA) keywords and word pacing.

The score is implemented to influence the user via visual feedback, encouraging the user to adapt their speech during the interaction while keeping the system’s behaviour simple and easy to interpret by the user [3].

The politeness score of each utterance will be displayed on the screen as neutral, polite, very polite, urgent and very urgent, while an on-screen timer keeps the pressure present.

	Gricean Maxims	Politeness Maxims
Primary use	Efficiency	Social cooperation
Speech Type	Concise + Direct	Indirect + Mitigation
Interaction Type	Fast + Blunt	Slower + Calmer

Code Snippet

```
1
2 def on_speech_input(audio):
3     text = recognise_speech(audio)
4     politeness = evaluate_fuzzy(text, pacing)
5
6     if politeness < THRESHOLD:
7         apply_penalty()
8     else:
9         increase_score(politeness)
10
11     update_screen_feedback()
12     display_score()
```

- Method**
- **Participants/Trials:** The user will play the game three times to observe adaptation.
 - **User Goal:** Move a ball around a triangular circuit using voice commands with a time limit.
 - **Input:** Speech is captured via microphone hardware and converted to text locally by VOSK.
 - **Affective computing:** A fuzzy logic system interprets linguistic cues and speech pacing to produce a proportional score.
 - **Scoring:** Politeness is weighted higher than pace, giving it priority. High scores are gained through polite and calm speech.
 - **Feedback:** On-screen timer give real-time visual feedback during the active game. The in-game screen displays the score for the previous utterance.
 - **Metric & analysis:** Final scores per round are compared across rounds to identify changes in interaction behaviour.

Soft Computing: Fuzzy Logic

$$\mu(x) = \begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a < x < b \\ 1 & x = b \\ \frac{c-x}{c-b} & b < x < c \\ 0 & x \geq c \end{cases}$$

Triangular membership functions are used to fuzzify words per second, while indirect speech act keywords are mapped to graded fuzzy values, both of which then contribute to a proportional politeness score [4].

UML Diagram

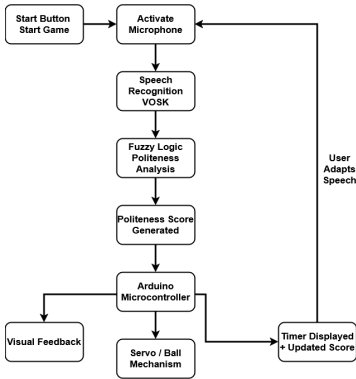


Figure 2: A UML activity diagram

Evaluation & Metrics

Users tended towards higher scores across repeated playthroughs, after initially feeling out the game mechanics and learning how to maximise phrase scores within the time limit. Improvements were primarily driven by more polite language and a slower pace of speech. This is consistent with findings on speech prosody and its effect on HRI [5]. Users learned which keywords and phrases netted more movement steps. Users were given minimal guidance beyond on-screen instructions and adapted their behaviour through real-time feedback rather than direct human instruction. A noticeable limitation was observed with the physical design of the artefact, long waiting times exist between interactions due to the player having to wait for the ball to reach its destination before it could move again.

Cybersecurity

The system was designed to avoid the storage of personal data. Users voice input is processed entirely locally by the open-source toolkit VOSK, with no external networking involved. Voice input is converted to text and immediately abstracted into a politeness score using a fuzzy logic algorithm. All raw data and processed data are discarded after use. Upon the completion of the game, the only data remaining is an anonymous score

derived from the ball completing circuits of the track. This adds another layer of abstraction on top of already non-identifiable data. This design prevents data from being repurposed outside the scope of the study, and the use of fuzzy logic rather than data-hungry learning systems improves real-time performance while retaining strong privacy safeguards.

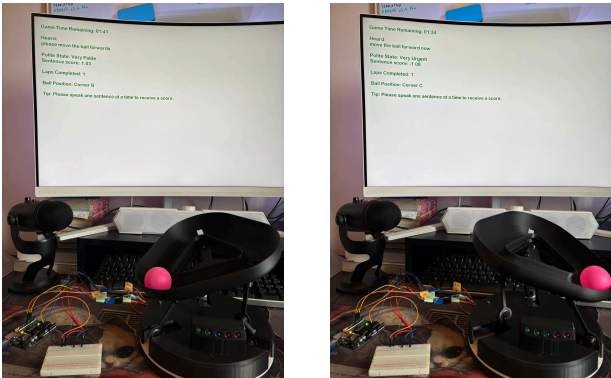


Figure 3: Game interface showing ball position and politeness-based score feedback during interaction.

Project Repository
GitHub Repository COMP304 Human-Robot Interaction

Bibliography

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